

MID Semester of 4th Sem B.Tech.

MID -SEMESTER EXAMINATION, FEBRUARY- MARCH, 2025

COURSE CODE: CBMTC405 & CIMTC405

COURSE TITLE: PROBABILITY THEORY AND STATISTICAL METHODS

TIME: 1.30 HRS

MAX. MARK: 20

Note:- Attempt all questions. Missing data/ information (if any), may be suitably assumed & mentioned in the answer.

Q. N	Question	Marks	CO												
1a	Let $p(x)$ be the probability function of discrete random variable X which assumes the values x_1, x_2, x_3, x_4 such that $2p(x_1) = 3p(x_2) = p(x_3) = 5p(x_4)$. Find the probability distribution and cumulative probability distribution of X .	2	CO1												
1b	An item is produced in large numbers. The machine is known to produce 5% defectives. A quality control inspector examines the items by taking them at random. What is the probability that at least 4 items are to be examined in order to get a defective product?	2	CO1												
2a	Daily consumption of milk in a city is 20,000 liters is approximately distributed as gamma variate with the parameters $\lambda = 2, \alpha = \frac{1}{10,000}$. The city has a daily stock of 30,000 liters. What is the probability stock is insufficient on a particular day?	2	CO1												
2b	Find the mean of the Hypergeometrical distribution.	2	CO1												
3a	Let X be a continuous random variable with pdf $f(x) = \begin{cases} ax, & 0 \leq x \leq 1 \\ a, & 1 \leq x \leq 2 \\ -ax + 3a, & 2 \leq x \leq 3 \\ 0, & \text{elsewhere} \end{cases}$. Compute $P(0.5 \leq X \leq 1.5)$.	2	CO1												
3b	Let variate X have the distribution $P(X = 0) = P(X = 2) = \alpha; P(X = 1) = 1 - 2\alpha$, for $0 \leq \alpha \leq \frac{1}{2}$. For what value of α is the $Var(X)$ a maximum?	2	CO2												
4a	Use Chebychev's inequality to determine how many times a fair coin must be tossed in order that the probability will be at least 0.95 that the ratio of the number of heads to the number of tosses will be between 0.45 and 0.55.	2	CO2												
4b	If the random variable X has the pdf $f(x) = \frac{x+1}{2}, -1 < x < 1$. Find the kurtosis of the random variable X .	2	CO2												
5a	The number of defects per unit in a sample of 330 units of manufactured products was found as follows: <table border="1"><tr><td>No of defects</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>No. of units</td><td>214</td><td>92</td><td>20</td><td>3</td><td>1</td></tr></table> Fit a Poisson distribution to the above-given data using mean $\lambda = 0.439$.	No of defects	0	1	2	3	4	No. of units	214	92	20	3	1	2	CO2
No of defects	0	1	2	3	4										
No. of units	214	92	20	3	1										
5b	Find the skewness of the random variable X with pdf $f(x) = \lambda e^{-\lambda x}, 0 \leq x < \infty, \lambda > 0$.	2	CO2												

Total No. of pages: 2

Roll No.

**B-TECH.- FOURTH SEMESTER (Backlog/Improvement)
MID-SEMESTER EXAMINATION, February 2025**

Course Code-COMTC13/CAMTC13/CBMTCT13/CDMTCT13/CIMTC13
Course Title-Probability and Stochastic Processes
Time: 1:30 hours
Max Marks-25

Note: Attempt all questions in given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO																		
1(a)	In a college, 25% of boys and 10% of girls are studying mathematics. The girls constitute 60% of the student body. If a mathematics student is selected at random, what's the probability that the student is (a) a girl, (b) a boy?	2.5	CO1																		
1(b)	A random variable X has the following distribution: <table border="1" style="margin: 10px auto;"><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>$P(x)$</td><td>0</td><td>k</td><td>$2k$</td><td>$3k$</td><td>k^2</td><td>$2k^2$</td><td>$7k^2$</td><td>k</td></tr></table> (i) Find k . (ii) Evaluate $P(X < 6)$, $P(X \geq 6)$ and $P(0 < X < 5)$.	x	0	1	2	3	4	5	6	7	$P(x)$	0	k	$2k$	$3k$	k^2	$2k^2$	$7k^2$	k	2.5	CO1
x	0	1	2	3	4	5	6	7													
$P(x)$	0	k	$2k$	$3k$	k^2	$2k^2$	$7k^2$	k													
2(a)	A probability curve $y = f(x)$ has a range from 0 to ∞ . If $f(x) = e^{-x}$, find the mean and the third moment about the mean.	2.5	CO1																		
2(b)	Define kurtosis of a probability distribution. Suppose the first four moments of a distribution are 1, 4, 10 and 46 respectively. Compute the coefficient of kurtosis and define its nature.	2.5	CO1																		
3(a)	Suppose a continuous random variable X has the probability density $f(x) = \begin{cases} k(1-x^2) & 0 < x < 1, \\ 0 & \text{otherwise.} \end{cases}$ (a) Find k . (b) Calculate the mean and variance.	2.5	CO2																		

3(b)	If X is having uniform distribution with mean 1 and variance $\frac{4}{3}$. Find $P(X < 0)$.	2.5	CO2
4(a)	After correcting 50 pages of a book, the proof reader finds that there are, on the average, 2 errors per 5 pages. How many pages would one expect to find with 0, 1, 2, 3 and 4 errors in 1000 pages of the first print of the book?	2.5	CO2
4(b)	A coin is weighted so that its probability of landing on heads is 20%. Suppose the coin is flipped 20 times. Find a bound for the probability that it lands on heads atleast 16 times.	2.5	CO2
5(a)	Out of 800 families with 5 children each, how many would you expect to have (a) 3 boys, (b) 5 girls, (c) either 2 or 3 boys?	2.5	CO2
5(b)	Define multinomial distribution in brief.	2.5	CO2